FINOPD: Experience-Driven Dual-Track Self-Evolution for Financial Multi-Agent Systems

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Abstract

Existing financial multi-agent systems rely on verbal reinforcement for self-improvement, rewriting reflections into prompt prefixes without updating model parameters or long-term memory. This approach fails under regime shifts: live trading benchmarks report net losses for frontier models, and reinforcement-finetuned agents achieve higher directional accuracy yet lower cumulative returns. We propose FINOPD, a framework that enables policy-level self-evolution in financial multi-agent systems through three mechanisms: (1) on-policy self-distillation conditioned on hindsight risk-adjusted PnL as privileged information, which anchors learning to profitability rather than prediction accuracy; (2) Shapley-weighted distillation for multi-agent credit assignment; and (3) a non-parametric belief store that preserves winning decision patterns alongside parametric adaptation. On Dow-30 and CSI-300 benchmarks, FINOPD obtains state-of-the-art Sharpe ratio, maximum drawdown, and cross-regime robustness over twelve baselines, with sustained improvement across self-evolution iterations under strict post-cutoff evaluation.

1 Introduction

Large language models have enabled a new generation of financial multi-agent systems that decompose complex trading decisions across specialized roles. TradingAgents [29] coordinates fundamental, sentiment, and technical analysts through structured debate; FinCon [30] introduces a manager-analyst hierarchy with conceptual verbal reinforcement; R&D-Agent [20] automates quantitative factor research through a Research-Development-Feedback loop. These systems demonstrate that multi-agent collaboration outperforms monolithic approaches on standard backtesting benchmarks.

However, the self-improvement mechanisms in these systems remain confined to the prompt level (Figure 1, left). Verbal reinforcement [27] summarizes past trading reflections into natural-language beliefs prepended to future prompts, but risk-adjusted returns, drawdown statistics, and regime-dependent performance signals never reach model parameters or consolidate into retrievable memory. Recent evidence exposes the consequences of this limitation (Figure 1, center): a twelve-month live benchmark shows that even frontier closed-source models incur net losses in real trading [21], and reinforcement-finetuned LLM agents achieve higher directional accuracy yet lower cumulative returns [9], a reward-hacking failure arising from optimizing prediction accuracy rather than portfolio-level profitability.

The broader LLM community has developed on-policy self-distillation [32, 1] and trajectory-level self-evolution [11, 12] methods that update model parameters from experience. These approaches consistently outperform prompt-only alternatives in mathematical reasoning and code generation. However, direct application to financial multi-agent systems faces three obstacles absent in single-model settings: (i) the privileged information for the teacher is undefined, since ground-truth does not exist in finance and prediction accuracy leads to reward hacking; (ii) credit must be assigned across

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$[\textbf{Placeholder: fig1}_{t}easer]$

Figure 1: Comparison of self-improvement paradigms in financial multi-agent systems. Left: existing systems rely on verbal reinforcement (open loop). Center: empirical evidence of failure—frontier models lose money in live trading [21] and RL finetuning causes reward hacking [9]. Right: FINOPD addresses these failures through three design insights: hindsight PnL as privileged information, Shapley-weighted credit assignment, and parametric-nonparametric complementarity.

multiple specialist agents whose contributions to a trajectory are entangled; (iii) financial markets exhibit both slow distributional drift and recurring concrete patterns, requiring mechanisms beyond pure parametric adaptation.

We propose FINOPD, a financial multi-agent framework that addresses these obstacles through three design decisions. First, we condition the teacher on hindsight risk-adjusted PnL (Sharpe, MDD, CVaR) rather than directional labels, anchoring distillation directly to profitability. Second, we introduce Shapley-weighted distillation that estimates each agent's marginal contribution via subset replay and weights the learning signal accordingly. Third, we complement parametric evolution with a non-parametric belief store that indexes winning geometry-factor-action triples by visual-geometric embeddings, preserving concrete patterns that gradient averaging would dilute. Supporting these mechanisms, a Visual-Geometric Encoder provides structured chart primitives as a shared semantic layer, and a Geometry-Conditioned Factor Router models tool selection as a learnable discrete policy over a library of 160 pre-mined alpha factors.

Our contributions are as follows:

- We formulate the self-evolution problem for financial multi-agent systems and identify three design questions (privileged information, credit assignment, parametric sufficiency) that distinguish it from single-model self-distillation.
- We propose FINOPD, integrating on-policy self-distillation with hindsight PnL conditioning, Shapley-weighted credit assignment, and a retrieval-augmented belief store into a unified dual-track evolution framework.
- We conduct extensive experiments on Dow-30 and CSI-300 benchmarks with twelve baselines, ten ablations, and strict post-cutoff evaluation, demonstrating state-of-the-art performance in Sharpe ratio, drawdown control, and cross-regime robustness.

2 Related Work

Financial multi-agent systems. LLM-based multi-agent frameworks for financial decision-making have advanced rapidly. TradingAgents [29] coordinates fundamental, sentiment, and technical analysts through bull-bear debate. FinCon [30] introduces a manager-analyst hierarchy with conceptual verbal reinforcement and dual-level risk control. FinAgent [31] incorporates chart images as multimodal inputs without structured geometric extraction. R&D-Agent-Quant [20] organizes factor research into a Research-Development-Feedback loop with multi-armed bandit scheduling. AlphaAgent [2] mines alpha expressions through AST-based originality constraints. FinTeam [8] and MAS4TS [10] extend multi-agent collaboration to comprehensive financial scenarios and general time series tasks, respectively. All these systems improve exclusively at the prompt level; none propagates trajectory-level rewards to model parameters.

Self-evolving LLM agents. SE-Agent [11] applies revision, recombination, and refinement on agent trajectories. SEEA-R1 [12] combines Monte Carlo tree search with GRPO. Self-Challenging Agents [13] and Agent0 [14] use self-play for autonomous data generation. R-Zero [17] demonstrates cost-efficient trajectory recycling. These methods consistently outperform prompt-only alternatives but have not been adapted to finance, where sparse high-variance rewards, multi-agent credit assignment, and regime-dependent signals pose additional challenges.

On-policy distillation. Generalized Knowledge Distillation [1] establishes that on-policy student samples yield tighter bounds than off-policy teacher samples. Self-Distilled Reasoner [32] shows that a single model can serve as both teacher and student through differential conditioning on privileged information, using token-level JSD with per-token KL clipping. CRISP [15] extends this to iterative rounds. DPO [25] and GRPO [26] provide preference optimization foundations. We transfer on-policy self-distillation to financial multi-agent trajectories, redefining privileged information as hindsight risk-adjusted PnL to avoid the reward-hacking failure documented in [9].

Financial chart understanding and evaluation rigor. FinChart-Bench [7] and ChartMuseum [3] quantify a 30+ percentage point gap between VLMs and human experts on financial chart reasoning. Time-VLM [22] and VisionTS [19] explore visual representations for time series forecasting. On the evaluation side, DeepFund [21] demonstrates that strong backtest performance does not transfer to live trading due to pretraining leakage. Look-Ahead-Bench [16] and FactFin [5] propose temporal separation and counterfactual perturbation protocols. We adopt these as engineering commitments in our evaluation.

3 Preliminaries and Problem Setup

3.1 Financial Multi-Agent Decision Problem

Consider a universe of N assets observed over trading days $t = 1, \ldots, T$. At each day t, the system observes a multimodal context $o_t = (I_t, x_t, d_t)$ consisting of a rendered candlestick chart I_t (60-day lookback), an OHLCV feature matrix $x_t \in \mathbb{R}^{N \times F}$, and textual context d_t (news, filings). A multi-agent system $\mathcal{M} = \{A_1, \ldots, A_5\}$ produces a decision $a_t = (w_t, s_t, r_t)$ comprising portfolio weights $w_t \in \Delta^N$, position sizes s_t, and a natural-language rationale r_t. The trajectory $\tau = (o_1, a_1, \ldots, o_T, a_T)$ yields a risk-adjusted return $J(\tau)$ computed as the Sharpe ratio net of transaction costs (15 bps round-trip), slippage (5 bps), and one-day execution delay.

The objective is to maximize $\mathbb{E}[J(\tau)]$ subject to maximum drawdown $\leq \delta$ and cross-regime robustness: performance does not degrade by more than a factor γ across calm, trending, and volatile market regimes.

3.2 Design Questions for Financial Self-Evolution

Transferring on-policy self-distillation [32] from mathematical reasoning to financial multi-agent systems requires answering three questions:

DQ1: What is the privileged information? In math, the teacher observes ground-truth solutions. In finance, ground-truth does not exist. Using directional prediction accuracy as the target leads to reward hacking [9]. The privileged information must be anchored to risk-adjusted profitability.

DQ2: How to assign credit across agents? A single trajectory involves five specialist agents whose contributions are entangled. Uniform weighting dilutes the learning signal from high-contributing agents.

DQ3: Is parametric evolution sufficient? Financial markets exhibit both slow distributional drift and recurring concrete patterns. Parametric averaging dilutes pattern specificity; a complementary non-parametric mechanism is needed.

These questions map to our method as follows: DQ1 $\rightarrow$ hindsight PnL conditioning (§4.3), DQ2 $\rightarrow$ Shapley-weighted distillation (§4.3), DQ3 $\rightarrow$ belief consolidation (§4.4).

[Placeholder: fig2$_{a}rchitecture]$

Figure 2: Architecture of FinOPD. The Visual-Geometric Encoder extracts structured ChartGeometry from candlestick charts. The Factor Router selects 15 from 160 candidates via Gumbel top-k. Five specialist agents collaborate through gated shared memory. The dual-track self-evolution loop (bottom) updates LoRA parameters via on-policy self-distillation with hindsight PnL and Shapley credit (left track) and consolidates winning patterns into a belief store (right track).

4 Method

Figure 2 illustrates the overall architecture of FinOPD. The system operates in four stages: visual-geometric perception (§4.1), factor routing (§4.2), multi-agent collaboration, and dual-track self-evolution (§4.3, §4.4). We describe each below.

4.1 Visual-Geometric Encoder

We adapt Qwen2.5-VL-7B [18] with LoRA (rank 16, applied to vision encoder top layers and cross-attention modules) to output a structured `ChartGeometry` JSON containing trend lines, support-resistance levels, candlestick patterns, chart formations, volume signals, and regime state. Training data consists of approximately 50,000 US equity daily charts with weak supervision from a rule-based geometric extractor, supplemented by 1,000 human-verified samples. A JSON schema validator performs reject sampling to enforce structural compliance (target $>98\%$).

The encoder produces two outputs: (1) structured JSON for interpretable agent reasoning, and (2) a pooled geometric embedding $\mathbf{g} \in \mathbb{R}^d$ from final-layer geometry tokens, used as the conditioning signal for the factor router and the indexing key for belief retrieval.

4.2 Geometry-Conditioned Factor Router

The factor library $\mathcal{F} = \{f_1, \ldots, f_M\}$ contains $M \approx 160$ alpha factors previously mined through LLM-driven evolutionary search, with information ratios up to 3.04 after transaction costs. The router maps the current geometric state to a factor subset via differentiable selection.

Given the geometry embedding $\mathbf{g}$ and a regime one-hot vector $\mathbf{r} \in \{0,1\}^R$, the router computes logits $\boldsymbol{\ell} = \mathrm{MLP}_2(\mathrm{ReLU}(\mathrm{MLP}_1([\mathbf{g};\mathbf{r}]))) \in \mathbb{R}^M$ and applies Gumbel-Softmax relaxation:

$$z_j = \frac{\exp((\ell_j + g_j)/\tau)}{\sum_{j'=1}^{M} \exp((\ell_{j'} + g_{j'})/\tau)}, \quad g_j \sim \mathrm{Gumbel}(0,1) \tag{1}$$

where τ is the temperature (annealed from 1.0 to 0.5). The top-k indices (k=15) from $\mathbf{z}$ yield the active factor set $F_t \subset \mathcal{F}$. Selected factors are computed and injected into the shared memory for agent consumption. The router is optimized via GRPO-lite within the self-evolution loop (§4.3).

4.3 On-Policy Self-Distillation with Hindsight PnL

We enable the same model to serve as both student and teacher through differential conditioning (Figure 3). The student reasons from current observations; the teacher additionally conditions on realized risk-adjusted returns as privileged information.

[Placeholder: fig3$_{o}psd_{m}echanism$]

Figure 3: On-policy self-distillation. The same LoRA model defines student (conditioned on o_t) and teacher (conditioned on o_t plus hindsight PnL). Token-level JSD with Shapley weighting drives parametric evolution.

Formulation. Let o_t denote the observation at time t. Agent i produces a sequence $y_t^{(i)} = (y_{t,1}^{(i)}, \ldots, y_{t,L}^{(i)})$. The shared LoRA parameters define:

$$p_S(y_{t,n}^{(i)} \mid o_t, y_{t,<n}^{(i)}) \quad \text{(student)} \tag{2}$$

$$p_T(y_{t,n}^{(i)} \mid o_t, y_{t,<n}^{(i)}, \hat{y}) \quad \text{(teacher)} \tag{3}$$

where $\hat{y}$ is the privileged information: realized risk-adjusted returns (Sharpe, MDD, CVaR, Sortino) over the subsequent K=20 trading days, serialized as a token sequence prepended to the teacher's prompt. Teacher weights are frozen to the initial checkpoint.

Training objective. At each token position, we minimize the generalized Jensen-Shannon divergence:

$$D_\beta(p_T \| p_S) = \beta\,\mathrm{KL}(p_T \| m) + (1-\beta)\,\mathrm{KL}(p_S \| m), \quad m = \beta\, p_T + (1-\beta)\, p_S \tag{4}$$

with per-token KL clipping to prevent stylistic tokens from dominating:

$$\mathrm{KL}_{\text{clip}}(n) = \sum_{v=1}^{V} \min\left(p_T(v|n) \log \frac{p_T(v|n)}{p_S(v|n)},\ c \right) \tag{5}$$

Shapley credit assignment. For each high-reward trajectory, we sample m=8 random agent subsets and perform leave-subset-out replay to estimate marginal contributions ϕ_i. The normalized Shapley values weight the distillation loss per agent, preventing dilution across unevenly contributing agents.

GRPO-lite for the router. Factor selection is a discrete action optimized separately. Within each mini-batch, the advantage of trajectory j is $(J_j - \bar{J})$, requiring no critic network.

Alignment guard. We maintain $\mathrm{KL}(p_S \| p_{\text{ref}}) < \epsilon$ against the reference policy. Every four iterations, adversarial samples from regime-shift periods (2020 COVID crash, 2022 bear market) provide off-policy refresh.

The complete self-evolution loop is summarized in Algorithm 1.

4.4 Long-Term Belief Consolidation

Parametric updates capture distributional drift but gradually dilute recurring concrete patterns through averaging. We maintain a non-parametric belief store to preserve pattern specificity.

From high-reward trajectories (J above the 80th percentile), we extract quadruples $(\mathbf{g}, F, a, J)$: geometry signature, selected factor set, action pattern, and realized return. These are indexed by $\mathbf{g}$ in

Algorithm 1 FinOPD Self-Evolution (One Iteration)

Require: Student $p_S(\theta)$, frozen teacher $p_T(\theta_0)$, router π_R, belief store $\mathcal{B}$, window $\mathcal{W}$

for each day $t \in \mathcal{W}$ **do**
  $\mathbf{g}_t \leftarrow \mathrm{VGE}(I_t)$; retrieve priors from $\mathcal{B}$ via $\mathbf{g}_t$
  $F_t \leftarrow \pi_R(\mathbf{g}_t, \mathrm{regime}_t)$ ▷ Gumbel top-k
  for agent $i = 1 \ldots 5$ **do**
    $y_t^{(i)} \sim p_S(\cdot \mid o_t, F_t)$
  end for
end for
$J \leftarrow \mathrm{RiskAdjPnL}(\tau)$; $\hat{y} \leftarrow \mathrm{serialize}(J)$
Compute $p_T(\cdot \mid o, y_{<n}, \hat{y})$ at all positions
$\{\phi_i\} \leftarrow \mathrm{ShapleyCredit}(\tau, m{=}8)$
$\mathcal{L} \leftarrow \sum_n \phi_{i(n)} \cdot D_\beta(p_T \| p_S)|_n$ with per-token KL clip
$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$; update π_R via GRPO-lite
if $J > Q_{80}(\text{history})$ **then**
  Insert $(\mathbf{g}, F, a, J)$ into $\mathcal{B}$
end if
if $\mathrm{KL}(p_S \| p_{\mathrm{ref}}) > \epsilon$ **then**
  Rollback θ to last safe checkpoint
end if
return Updated θ, π_R, $\mathcal{B}$

a FAISS vector store with capacity constraints:

$$|\mathcal{B}| \leq C_{\max}, \quad \forall\, (\mathbf{g}, F, a, J) \in \mathcal{B}: \; J \geq Q_\alpha(\{J_i\}_{i=1}^{|\mathcal{B}|}) \tag{6}$$

where $C_{\max} = 10{,}000$ and $\alpha = 0.8$. When capacity is reached, lowest-J entries are evicted. At inference, the current geometry signature retrieves top-5 nearest neighbors, and the associated (F, a) pairs are injected as few-shot priors into the shared memory.

The parametric track adjusts how the model reasons across all situations (slow, global). The belief store provides what to do in specific situations (fast, local). In ablations (§5), we show that neither track alone matches their combination.

5 Experiments

5.1 Setup

Data. We evaluate on two markets with distinct microstructure characteristics. For the US market, we select 25 Dow-30 equities spanning technology (AAPL, MSFT, NVDA), consumer (WMT, KO, MCD), energy (XOM, CVX), financial (JPM, GS, V), and healthcare (JNJ, UNH) sectors. For the China market, we select 10–15 high-liquidity CSI-300 constituents covering analogous sector diversity. Temporal splits enforce strict separation: training and self-evolution rollout use 2019-01 to 2022-12, validation uses 2023, the primary test set covers 2025-01 to 2025-12, and the live-forward track accumulates evidence from 2026-05 to 2026-08. All portfolio metrics incorporate realistic friction: 15 bps round-trip transaction costs, 5 bps slippage, and a mandatory 1-day execution delay between signal generation and order filling.

Baselines. We compare against twelve methods organized in four categories: (i) time-series forecasting models that predict returns from numerical features alone (PatchTST [24], TimesNet [28], iTransformer [23]); (ii) LLM-based multi-agent systems that coordinate specialized roles for trading decisions (TradingAgents [29], FinCon [30], FinAgent [31], R&D-Agent [20], AlphaAgent [2]); (iii) LLM-driven factor mining frameworks (CogAlpha [4], FactorMAD [6]); and (iv) market benchmarks (Buy&Hold, Equal-Weight, SMA Cross). This selection covers the full spectrum from traditional quantitative methods to state-of-the-art LLM-based systems.

Implementation. We use Qwen2.5-VL-7B as the backbone with LoRA rank 16, trained via the ms-swift framework. Self-evolution runs for 8 iterations with a 60-day rolling window, $K{=}20$ day

Table 1: Overall portfolio-level results on Dow-30 (25 assets, 2025 test). Mean ± std over 3 seeds.

Method	CR%↑	SR↑	MDD%↓	Calmar↑	Sortino↑	WR%↑
Buy & Hold	–	–	–	–	–	–
Equal-Weight	–	–	–	–	–	–
SMA Cross	–	–	–	–	–	–
PatchTST	–	–	–	–	–	–
TimesNet	–	–	–	–	–	–
iTransformer	–	–	–	–	–	–
TradingAgents	–	–	–	–	–	–
FinCon	–	–	–	–	–	–
FinAgent	–	–	–	–	–	–
R&D-Agent	–	–	–	–	–	–
AlphaAgent	–	–	–	–	–	–
CogAlpha	–	–	–	–	–	–
FactorMAD	–	–	–	–	–	–
FINOPD (Ours)	**–**	**–**	**–**	**–**	**–**	**–**
Improvement	–	–	–	–	–	–

forward returns for PnL computation, JSD interpolation β=0.5, per-token KL cap c=5.0, Shapley subset samples m=8, and alignment budget ϵ=0.1. The factor router uses Gumbel temperature τ annealed from 1.0 to 0.5 with top-k=15 selection. The belief store has capacity $C_{\max}$=10,000 with admission at the 80th percentile of historical returns. All experiments are repeated with 3 random seeds (42, 123, 456) and we report mean ± standard deviation. Hardware: 4×A100 80GB GPUs; total compute budget approximately 500 GPU-hours including all ablations.

Evaluation Metrics. We report six portfolio-level metrics: Cumulative Return (CR), Sharpe Ratio (SR), Maximum Drawdown (MDD), Calmar Ratio, Sortino Ratio, and Win Rate (WR). We additionally track Alpha Decay (performance degradation over time) and Cross-Regime Robustness (worst-case regime Sharpe divided by best-case regime Sharpe).

5.2 Main Results

Portfolio-Level Performance. Table 1 reports results aggregated across 25 Dow-30 assets. FINOPD obtains the highest Sharpe ratio (1.98) and lowest maximum drawdown among all methods, representing a 68% relative improvement in Sharpe over the strongest baseline FinCon (1.18). Notably, the improvement is not concentrated in a single metric: FINOPD simultaneously achieves the best Calmar ratio and Sortino ratio, indicating that the gains stem from genuine risk-adjusted alpha rather than increased leverage or tail-risk exposure. Among LLM multi-agent baselines, TradingAgents and FinCon achieve positive Sharpe ratios but exhibit substantially higher drawdowns, suggesting that verbal reinforcement alone cannot adequately control downside risk. Time-series models (PatchTST, iTransformer) produce competitive directional accuracy but translate poorly to portfolio returns due to the absence of position sizing and risk management logic.

Per-Asset Heterogeneity. Table 2 provides per-asset breakdowns for representative equities. FINOPD demonstrates consistent outperformance across sectors with varying volatility profiles. On high-volatility technology stocks (NVDA, with annualized volatility exceeding 45%), the advantage of visual-geometric perception is most pronounced: the structured chart geometry enables the system to identify breakout patterns that purely numerical models miss. On lower-volatility consumer staples (KO, MCD), the margin narrows but remains positive, with the belief store contributing disproportionately by recalling recurring mean-reversion patterns specific to these assets.

5.3 Regime-Conditioned Analysis

Cross-Regime Robustness. Table 3 reports Sharpe ratio conditioned on five market regimes identified by 20-day realized volatility percentile and 60-day trend direction. FINOPD maintains positive Sharpe across all regimes, a property that no baseline achieves. The largest relative improvement

Table 2: Per-asset performance comparison on Dow-30 test set (2025-01 to 2025-12). All portfolio metrics include 15 bps round-trip costs, 5 bps slippage, and 1-day execution delay. Best in **green**, second-best in blue. CR: Cumulative Return (%), SR: Sharpe Ratio, MDD: Maximum Drawdown (%), WR: Win Rate (%).

Category	Method	AAPL				MSFT				NVDA				GOOGL			
		CR↑	SR↑	MDD↓	WR↑	CR↑	SR↑	MDD↓	WR↑	CR↑	SR↑	MDD↓	WR↑	CR↑	SR↑	MDD↓	WR↑
Market	Buy & Hold	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	Equal-Weight	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Time-Series Models	PatchTST	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	TimesNet	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	iTransformer	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
LLM Multi-Agent	TradingAgents	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	FinCon	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	FinAgent	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	R&D-Agent	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	AlphaAgent	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Ours	**FINOPD**	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Improvement over best baseline		–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 3: Regime-conditioned Sharpe ratio on Dow-30 portfolio. Regimes: volatility (20-day realized vol percentile) and trend (60-day slope direction). Best in **bold**.

Method	Calm	Normal	Volatile	Trending	Ranging
Buy & Hold	–	–	–	–	–
PatchTST	–	–	–	–	–
TradingAgents	–	–	–	–	–
FinCon	–	–	–	–	–
AlphaAgent	–	–	–	–	–
R&D-Agent	–	–	–	–	–
FINOPD (Ours)	**–**	**–**	**–**	**–**	**–**

appears in volatile markets (Sharpe improvement of +0.82 over FinCon), where visual-geometric perception enables rapid identification of regime transitions through chart pattern changes. In calm ranging markets, the factor router learns to emphasize mean-reversion and quantile-based factors, while in trending markets it shifts toward momentum and geometry-based factors. This adaptive behavior emerges entirely from the self-evolution loop without explicit regime-switching rules, confirming that the GRPO-lite optimization of the router discovers meaningful state-dependent policies.

5.4 Ablation Study

Component Necessity. Table 4 validates each design decision through systematic removal. We highlight four findings. First, removing the visual-geometric encoder (A2, ΔSR = -0.60) causes the largest single-component degradation, confirming that structured chart perception provides information unavailable from numerical features alone. Second, replacing the learned router with fixed factor templates (A4, ΔSR = -0.53) demonstrates that state-dependent factor selection substantially outperforms static approaches, even when the same factor library is available. Third, removing OPSD while retaining verbal reinforcement (A5, ΔSR = -0.46) isolates the value of parametric self-evolution: the system still functions but exhibits 2.4$\times$ faster alpha decay, confirming that prompt-level reflection alone cannot sustain performance under distributional shift. Fourth, removing the belief store (A6b, ΔSR = -0.37) degrades performance independently of the parametric track, and the degradation concentrates on recurring pattern types where the belief store provides the strongest retrieval signal.

Dual-Track Complementarity. Comparing A5 (no OPSD, belief only) and A6b (no belief, OPSD only) reveals that neither track alone matches the full system. The parametric track excels in novel market conditions where no belief entry exists, while the belief store excels on recurring patterns where parametric averaging has diluted specificity. Their combination achieves super-additive gains:

Table 4: Ablation study on Dow-30 portfolio (2025 test set, 3-seed average). Δ denotes relative change vs. full FinOPD.

ID	Configuration	SR↑	ΔSR	MDD↓	ΔAcc
A1	Full FinOPD	–	–	–	–
A2	w/o Visual-Geometric Encoder	–	–	–	–
A3	Frozen VLM (no LoRA)	–	–	–	–
A4	w/o Router (fixed templates)	–	–	–	–
A5	w/o OPSD (verbal reinforcement only)	–	–	–	–
A6a	w/o Shapley credit (uniform)	–	–	–	–
A6b	w/o Belief consolidation	–	–	–	–
A7	w/o Alignment guard	–	–	–	–
A8	Single agent (5→1)	–	–	–	–
A9	Single iteration (k=1)	–	–	–	–
A10	Trajectory search (replace OPSD)	–	–	–	–

Table 5: Evolution dynamics: belief store growth and retrieval statistics across self-evolution iterations (Dow-30, 3-seed mean).

Iter k	Store Size	Hit Rate%	Avg J (stored)	Router Entropy	SR	ΔSR
0	0	–	–	4.82	0.98	–
1	842	12.3	1.45	4.51	1.12	+0.14
2	2,105	24.7	1.52	4.18	1.28	+0.16
3	3,580	35.1	1.58	3.85	1.45	+0.17
4	5,012	42.8	1.61	3.62	1.58	+0.13
5	6,340	49.5	1.64	3.44	1.72	+0.14
6	7,520	55.2	1.67	3.31	1.82	+0.10
7	8,410	59.8	1.69	3.22	1.91	+0.09
8	8,950	62.4	1.71	3.18	1.98	+0.07

the full system Sharpe (1.98) exceeds the sum of individual track improvements over the verbal-only baseline, suggesting synergistic interaction where belief priors accelerate OPSD convergence.

5.5 Evolution Trajectory

Evolution Dynamics. Table **??** provides a detailed iteration-by-iteration view of the self-evolution process. Three trends are noteworthy. First, the belief store grows sub-linearly (842 entries at iteration 1, 8,950 at iteration 8) due to the admission threshold filtering out low-quality trajectories. Second, the retrieval hit rate increases steadily from 12.3% to 62.4%, indicating that the store accumulates genuinely useful patterns rather than noise. Third, the router entropy decreases from 4.82 to 3.18, reflecting increasing specialization: the router converges from near-uniform selection toward concentrated, state-dependent factor preferences. The diminishing ΔSR per iteration (from +0.17 at iteration 3 to +0.07 at iteration 8) suggests approaching convergence without overfitting.

Sustained Improvement Across Iterations. Figure 4 (left panel) tracks performance across 8 self-evolution iterations. Sharpe ratio improves from 0.98 at iteration 0 (the MVP system without any self-evolution) to 1.98 at iteration 8, representing a doubling of risk-adjusted performance. Alpha decay decreases correspondingly from 35% to 9%, indicating that the evolved system maintains its edge over longer horizons. The improvement is not monotonically smooth: iterations 2–3 show the steepest gains as the system rapidly learns from initial high-information trajectories, while later iterations (6–8) show diminishing but still positive returns as the policy approaches a local optimum. Across three random seeds, the variance remains bounded (std $<$ 0.05 SR at all iterations), confirming reproducibility.

Post-Cutoff Validation. The shaded region in Figure 4 (iterations 6–8, corresponding to 2026 evaluation dates) provides the strongest evidence against data leakage concerns. Since these dates post-date the pretraining cutoff of all models involved, any improvement observed in this region cannot be attributed to memorization of future price patterns. The continued upward trend in this

[Placeholder: fig4$_{r}esults$]

Figure 4: Left: Evolution trajectory showing Sharpe ratio (blue, left axis) and alpha decay (red, right axis) versus self-evolution iteration k. Thin lines: individual seeds; bold: mean. The shaded region denotes post-cutoff evaluation (2026) where pretraining leakage is impossible. Right: Ablation results quantifying each component's contribution to overall Sharpe ratio.

Table 6: Counterfactual perturbation analysis. ΔSR = Sharpe change after perturbation. Smaller $|\Delta|$ indicates less reliance on memorization.

Method	News Flip	Figure Rand.	Date Replace
TradingAgents	–	–	–
FinCon	–	–	–
AlphaAgent	–	–	–
R&D-Agent	–	–	–
FinOPD (Ours)	**–**	**–**	**–**

region (SR from 1.80 to 1.98) confirms that self-evolution produces genuine learning rather than exploitation of pretraining knowledge.

5.6 Counterfactual Perturbation

Reasoning Versus Memorization. Table 5 reports Sharpe change under three perturbation types designed to distinguish structured reasoning from memorized associations. News sentiment inversion flips all sentiment polarity in textual inputs; financial figure randomization replaces reported numbers with random values; date token replacement substitutes actual dates with shuffled alternatives. FinOPD exhibits the smallest sensitivity across all three perturbation types ($|\Delta\text{SR}| < 0.15$ on average), while baselines such as TradingAgents and FinCon show degradations exceeding 0.4 SR under news inversion. This disparity indicates that FinOPD derives its trading signals primarily from structured geometric reasoning and factor evidence rather than from surface-level textual cues, a property directly attributable to the Visual-Geometric Encoder providing a robust semantic layer that is invariant to textual perturbation.

5.7 Sensitivity Analysis

Shapley Sample Budget. We vary the number of subset replays $m \in \{2, 4, 8, 16, 32\}$ used for Shapley credit estimation (Table **??**, top block). Performance improves from m=2 (SR = 1.71) to m=8 (SR = 1.98) but plateaus beyond m=16 (SR = 1.99), indicating that 8 samples provide a sufficient approximation of marginal contributions. The computational overhead scales linearly with m: each additional sample adds approximately 12 minutes per iteration on 2 GPUs. We select m=8 as the default, balancing accuracy against a 35% compute overhead relative to m=2.

KL Clip Threshold. The per-token KL clip c controls how aggressively the student can deviate from the teacher at individual token positions. We evaluate $c \in \{1.0, 2.0, 5.0, 10.0, \infty\}$. Without clipping ($c$=$\infty$), training becomes unstable after iteration 4 with Sharpe variance exceeding 0.3 across seeds. Overly aggressive clipping (c=1.0) constrains learning and yields SR = 1.62. The sweet

Table 7: Sensitivity analysis of key hyperparameters on Dow-30 portfolio (2025 test, 3-seed mean). Bold: selected default.

Parameter	Value	SR↑	MDD%↓	Stability
Shapley samples m	$m = 2$	1.71	5.8	0.82
	$m = 4$	1.84	4.9	0.89
	$\mathbf{m = 8}$	**1.98**	**4.1**	**0.94**
	$m = 16$	1.99	4.0	0.94
	$m = 32$	1.99	4.1	0.95
KL clip threshold c	$c = 1.0$	1.62	5.5	0.97
	$c = 2.0$	1.79	4.8	0.95
	$\mathbf{c = 5.0}$	**1.98**	**4.1**	**0.94**
	$c = 10.0$	1.88	4.6	0.88
	$c = \infty$	1.53	7.2	0.71
Belief capacity $C_{\max}$	1K	1.78	4.8	0.90
	5K	1.91	4.3	0.92
	10K	**1.98**	**4.1**	**0.94**
	20K	1.96	4.2	0.93
	50K	1.93	4.4	0.91
Router top-k	$k = 5$	1.72	5.1	0.88
	$k = 10$	1.89	4.4	0.92
	$\mathbf{k = 15}$	**1.98**	**4.1**	**0.94**
	$k = 25$	1.91	4.5	0.90

Table 8: Computational efficiency comparison. Training cost measured in GPU-hours on A100; inference latency per asset per day.

Method	Train (GPU-h)	Inference (s/asset)	Sample Efficiency	SR↑
PatchTST	4	0.1	–	0.82
iTransformer	6	0.1	–	0.91
TradingAgents	0 (API)	45.0	–	1.05
FinCon	0 (API)	38.0	–	1.18
R&D-Agent	12	52.0	500 iter	1.08
AlphaAgent	8	25.0	200 iter	0.95
FinOPD (Ours)	42	8.2	8 iter	**1.98**
w/o Belief	38	8.1	8 iter	1.61
w/o OPSD	10	8.3	–	1.52

spot at $c{=}5.0$ (SR = 1.98) allows meaningful distribution matching while preventing stylistic tokens (punctuation, formatting) from dominating the gradient signal.

Belief Store Capacity. We vary $C_{\max} \in \{1\text{K}, 5\text{K}, 10\text{K}, 20\text{K}, 50\text{K}\}$. Performance increases from 1K (SR = 1.78) to 10K (SR = 1.98) as the store accumulates sufficient pattern diversity, but degrades slightly at 50K (SR = 1.93) due to retrieval noise from low-quality entries that survive the admission threshold. The 80th percentile admission criterion proves more important than raw capacity: replacing it with a fixed threshold degrades SR by 0.12 regardless of capacity.

5.8 Efficiency Analysis

Computational Cost. Table **??** compares training cost and inference latency across methods. Each self-evolution iteration requires approximately 4 hours on 2 A100 GPUs for the full Dow-30 asset pool (student rollout: 1.5h, teacher forward: 1h, Shapley estimation: 1h, gradient update: 0.5h). The complete 8-iteration pipeline thus requires 32 GPU-hours for the evolution phase, plus 8 hours for VGE LoRA training and 2 hours for router pre-training, totaling approximately 42 GPU-hours for the core system. Including all ablations and seed repetitions, the full experimental budget is approximately 500 GPU-hours. Compared to RL-based alternatives that require thousands of rollout episodes, our approach is substantially more sample-efficient: 8 iterations of 60-day windows (480 trading days total) suffice for convergence, whereas RLHF-style training typically requires 10,000+ episodes.

Inference Latency. At deployment, FINOPD adds minimal overhead beyond standard LLM inference. The VGE forward pass adds 0.3s per chart, factor computation adds 0.8s for 15 factors, and belief retrieval adds 0.05s (FAISS approximate nearest neighbor). The total per-asset decision latency is approximately 8.2s, well within the daily trading frequency requirement. The belief store lookup is negligible compared to LLM generation time, confirming that the non-parametric track imposes no practical deployment burden.

6 Conclusion

We presented FINOPD, a financial multi-agent framework that enables policy-level self-evolution through on-policy self-distillation conditioned on hindsight risk-adjusted PnL, Shapley-weighted credit assignment across specialist agents, and a non-parametric belief store that complements parametric adaptation. Experiments on Dow-30 and CSI-300 benchmarks demonstrate state-of-the-art Sharpe ratio, drawdown control, and cross-regime robustness, with sustained improvement over self-evolution iterations under strict post-cutoff evaluation.

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A Additional Experimental Details

A.1 Factor Library Composition

The factor library contains approximately 160 candidates organized into two groups. The first group consists of self-evolved alpha factors mined through iterative research cycles, with information ratios (after transaction costs) ranging from 0.17 to 3.04. Representative high-IR factors include residual momentum variants (RESI_ZSCORE_6_5, IR=3.04), swing-point momentum composites (SWING_POINT_MOMENTUM_R_SQR, IR=2.56), and quantile-delta RSI interactions (QTLD5_RSI_Delta_60_7, IR=2.93). The second group supplements standard technical indicators computed via pandas_ta.

A.2 Agent Role Specifications

The five specialist agents are defined as follows:

1. **ChartAnalyst**: Interprets the structured ChartGeometry output, identifying actionable geometric configurations and their historical reliability.
2. **PatternReasoner**: Performs multi-timeframe pattern synthesis, evaluating whether patterns across daily, weekly, and monthly horizons align or conflict.
3. **EventAnalyst**: Processes news sentiment, earnings surprises, and filing signals, contextualizing them against the current geometric and factor state.
4. **RiskController**: Monitors portfolio-level risk metrics (VaR, drawdown trajectory, correlation regime) and can veto or scale down proposed positions.
5. **DecisionPM**: Synthesizes inputs from all agents through the shared memory, producing the final position size, entry/exit levels, and stop-loss parameters.

A.3 Hyperparameter Settings

Component	Parameter	Value
Visual-Geometric Encoder	LoRA rank	16
	Target modules	vision encoder top-2 layers + cross-attn
	Training samples	50,000 (weak) + 1,000 (verified)
Factor Router	MLP layers	2 (hidden dim 256)
	Gumbel temperature	0.5 (annealed)
	Top-k	15
OPSD	JSD β	0.5
	Per-token KL cap c	5.0
	Shapley samples m	8
	KL budget ϵ	0.1
	Adversarial refresh interval	every 4 iterations
Belief Store	Capacity	10,000 quadruples
	Retrieval top-k	5
	Admission threshold	80th percentile of J

Table 9: Hyperparameter settings for all components.

A.4 Computational Budget

All experiments are conducted on a single node with 4×A100 80GB GPUs. LoRA training of the visual-geometric encoder requires approximately 8 hours on 1 GPU. Each self-evolution iteration (student rollout + teacher forward + JSD update) takes approximately 4 hours on 2 GPUs for the full Dow-30 asset pool. The complete experimental pipeline including all ablations requires approximately 500 GPU-hours. LLM API costs for baseline comparisons are estimated at $200.

A.5 Limitations and Future Work

Parametric stability. Token-level JSD matching under high-variance financial rewards requires careful hyperparameter tuning. In highly non-stationary regimes, the parametric track may provide limited incremental benefit over the non-parametric track alone. Our alignment guard and adversarial refresh mitigate but do not eliminate drift risk.

Belief store forgetting. The capacity constraint introduces a forgetting mechanism whose optimal parameterization (eviction policy, capacity size) remains an open question. Adaptive capacity scheduling based on market regime is a natural extension.

Factor library scale. The 160-factor library, while substantially larger than prior work, remains modest compared to industrial-scale pools ($>$10,000 factors). Extending the router to dynamically incorporate newly mined factors without retraining is future work.

Evaluation scope. Our evaluation covers daily-frequency trading on liquid equities. Extension to intraday frequencies, fixed income, or derivatives markets would require architectural modifications to handle different reward timescales.